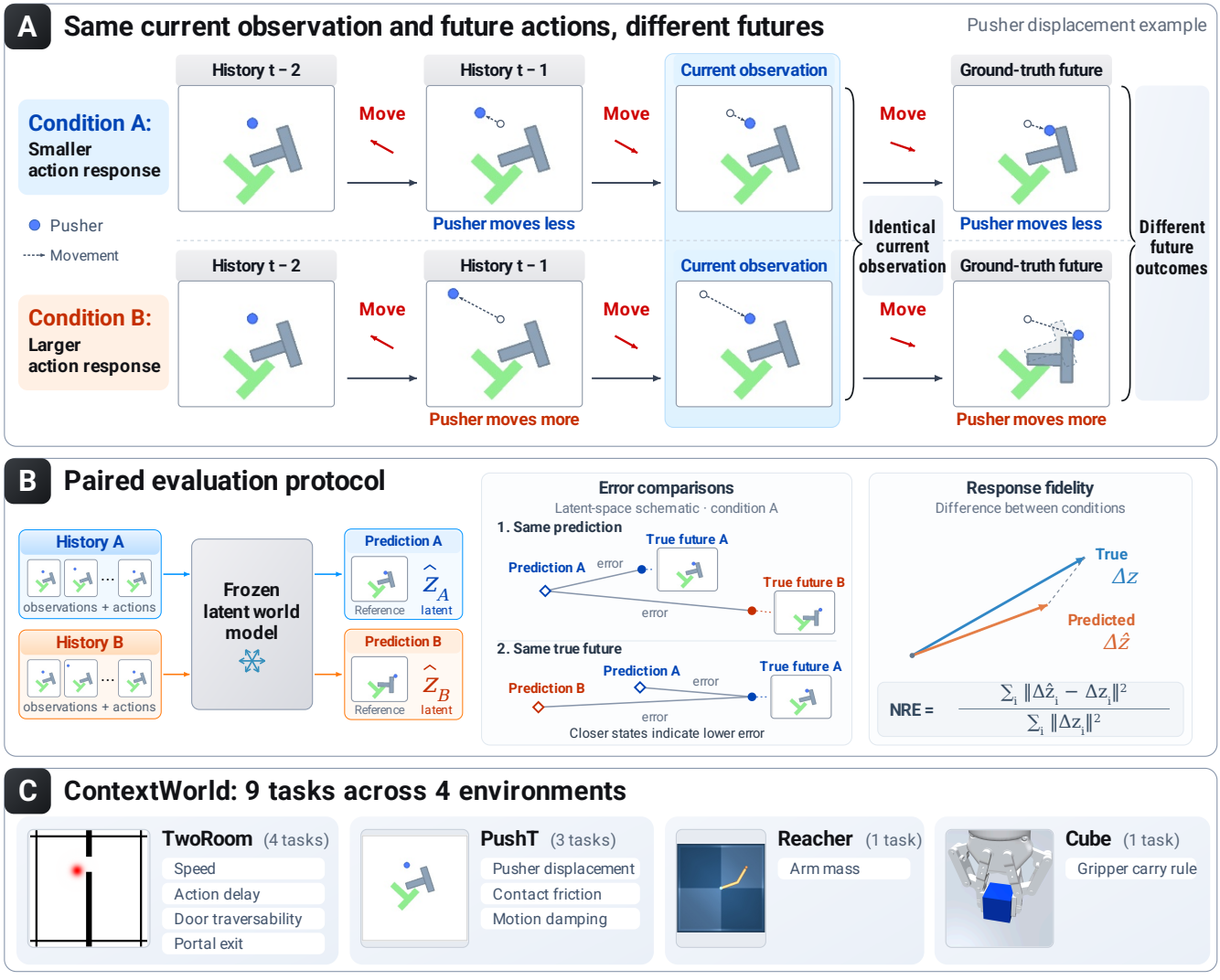

A
Same current observation and future actions, different futures
Pusher displacement example
Condition A:
Smaller
action response
Pusher
Movement
Condition B:
Larger
action response
History t − 2
History t − 1
Current observation
Ground-truth future
Move
Pusher moves less
Pusher moves more
Identical
current
observation
Different
future
outcomes
B
Paired evaluation protocol
History A
observations + actions
History B
observations + actions
Frozen
latent world
model
Prediction A
Prediction B
Reference
latent
Error comparisons
Latent-space schematic · condition A
1. Same prediction
Prediction A
error
True future A
True future B
2. Same true future
Prediction B
Prediction A
True future A
Closer states indicate lower error
Response fidelity
Difference between conditions
True
Predicted
NRE =
C
ContextWorld: 9 tasks across 4 environments
TwoRoom (4 tasks)
Speed
Action delay
Door traversability
Portal exit
PushT (3 tasks)
Pusher displacement
Contact friction
Motion damping
Reacher (1 task)
Arm mass
Cube (1 task)
Gripper carry rule